

Lab 1 - Integration

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```
myFunction=function(x){ x^2+3*x-4}
print(myFunction)

## function (x)
## {
##      x^2 + 3 * x - 4
## }

myFunction(0)

## [1] -4
myFunction(1)

## [1] 0
myFunction(100)

## [1] 10296
integrate(myFunction, 0,2)

## 0.6666667 with absolute error < 5.5e-14
#1st
whatthefunc=function(x){exp(3*x^2)*x^3}
integrate(whatthefunc, -0.5,1)

## 2.261129 with absolute error < 4.7e-09
#2nd
funcmylife=function(x){((x^3)/(sin(x))^.5)}
integrate(funcmylife, 0,pi)

## 51.73292 with absolute error < 0.0016
#3rd
funcoff=function(x){(exp((-1*x^2)/2)/(2*pi))}
integrate(funcoff, -1,1)

## 0.2723537 with absolute error < 3e-15
#4th
holysfunc=function(x){(exp((-1*x^2)/2)/(2*pi))}
integrate(holysfunc, -2,2)

## 0.3807903 with absolute error < 7.3e-12
```

```
#5h
funcingA=function(x){(exp((-1*x^2)/2)/(2*pi))}
integrate(funcingA, -3,3)
```

```
## 0.3978652 with absolute error < 3.7e-07
```

Double Integration

```
require(cubature)
```

```
## Loading required package: cubature
```

```
## Warning: package 'cubature' was built under R version 4.4.3
```

```
myFunctionD=function(vec){ vec[1]^2+3*vec[1]*vec[2]-4*vec[2]^3}
print(myFunctionD)
```

```
## function (vec)
## {
##     vec[1]^2 + 3 * vec[1] * vec[2] - 4 * vec[2]^3
## }
```

```
adaptIntegrate(myFunctionD, c(0,1), c(2,1.5))
```

```
## $integral
## [1] -3.041667
##
## $error
## [1] 0
##
## $functionEvaluations
## [1] 17
##
## $returnCode
## [1] 0
```

```
#1st
whatthefuncd=function(vec){ cos(vec[1])*(vec[2])^2}
adaptIntegrate(whatthefuncd, c(1,2), c(0,pi/2))
```

```
## $integral
## [1] 1.156803
##
## $error
## [1] 8.600794e-07
##
## $functionEvaluations
## [1] 17
##
## $returnCode
## [1] 0
```

```
#2nd
funcmylifed=function(vec){exp(vec[1]^2)*2^(vec[2])}
adaptIntegrate(funcmylifed, c(0,.1), c(-2,2))
```

```
## $integral
```

```
## [1] -69.50475
##
## $error
## [1] 0.0005475603
##
## $functionEvaluations
## [1] 357
##
## $returnCode
## [1] 0

#3rd
funcoffd=function(vec){(2*pi)^-1*exp(-2^-1*(vec[1]^2+vec[2]))}
adaptIntegrate(funcoffd, c(-2,2), c(-2,2))

## $integral
## [1] 0
##
## $error
## [1] 0
##
## $functionEvaluations
## [1] 17
##
## $returnCode
## [1] 0
```